Validation did not confirm the original selections

Report 2 of 3 · Validation and frozen selection · Consolidated final edition

Executive Summary

- **All four development-selected candidates lost in 2024–2025**, with net returns from **–22.31% to –45.31%** at 20 bps. Their positive development Sharpes did not generalize to the next two years.
- The broad development-Q1 group retained a useful relative ordering, but not positive median economics: its validation median return was **–16.27%** and **30.73%** of cases were profitable. “Q1” is the best approximately 20% of development Sharpe ranks within each basket, not a calendar quarter.
- A separate validation-informed rule produced **five Track B candidates** from **39,332 eligible pairs**. Their validation Sharpes of **1.14–1.85 are selection statistics**, not independent evidence of discovery. B02 had no eligible candidate.
- The final research shortlist preserves the **original four plus five Track B selections**. No single combined portfolio, weights or winner is implied.

Read the periods correctly

Development: 1 January 2022–31 December 2023. Design and diagnose the rule using this period only. **Validation: 1 January 2024–31 December 2025.** Test Track A unchanged, then explicitly label later validation-informed selection as Track B.

The reserved evaluation was subsequently approved for **1 January–30 September 2026**, with an October 1 UTC closing endpoint. Its outcomes belong only in Report 3. This edition reads saved accepted research; it does not load prices, rerun strategies or change any selection.

Primary convention: 20 bps one-way fees on full-L1 target changes. Lag1 means idealized observed-close execution, not a real one-hour delay.

1. Track A tested development robustness, not the validation winner

The original development-only screen required positive 2022 and 2023 returns and Sharpes, positive full-development returns under 40 bps and an additional-hour execution delay, and more than half of both parameter and all 28 one-swap basket neighbors to be profitable.

The robustness score was the minimum of the two annual Sharpes and the two neighborhood median Sharpes. At most one case-basket was selected per family; score ties used lower turnover, experiment ID and basket ID. No family gate was relaxed when empty.

The rule was designed after development robustness had been inspected. Freezing it prevents later reselection; it does not remove development search bias or earlier historical inspection.

The same four identities in both periods

ID	Period	Net return	Sharpe	Annual L1 turnover	Max drawdown
A01	Development	+165.36%	1.84	31.44×	23.25%
A01	Validation	-29.55%	-0.39	35.15×	44.82%
A02	Development	+252.94%	2.03	49.58×	58.37%
A02	Validation	-39.63%	-0.64	55.43×	47.29%
A03	Development	+64.38%	1.19	27.62×	19.79%
A03	Validation	-22.31%	-0.39	22.34×	43.11%
A04	Development	+314.22%	1.43	50.50×	51.35%
A04	Validation	-45.31%	-0.22	63.91×	70.96%

All four failed the economic expectation of positive net validation returns. The result is not simply that they failed to remain rank one. Original risk also persisted: A04's validation maximum drawdown reached **70.96%**. The four stay in the record unchanged rather than being replaced by favorable controls.

2. The reference controls isolate a basket comparison

“Reference” means the **predefined liquidity-tier reference basket**, ID 0: BTC, ETH, SHIB, BNB, AVAX, DOGE, FIL and GALA. It is a fixed comparison, not a claim that this basket is optimal.

Each row compares exactly the same strategy settings on the selected basket and reference basket. The selected baskets had much stronger development results. In validation, reference momentum and long/short trend were slightly profitable, while their selected-basket versions lost. Reversal was worse on the reference basket, so there is no universal reference advantage.

Same settings, selected versus reference basket

ID	Period	Selected net	Reference net	Selected Sharpe	Reference Sharpe
A01	Development	+165.36%	+8.19%	1.84	0.28
A01	Validation	-29.55%	-9.90%	-0.39	-0.06
A02	Development	+252.94%	-13.26%	2.03	-0.10
A02	Validation	-39.63%	+8.66%	-0.64	0.29
A03	Development	+64.38%	+12.63%	1.19	0.35
A03	Validation	-22.31%	-38.60%	-0.39	-0.73
A04	Development	+314.22%	+98.46%	1.43	0.86
A04	Validation	-45.31%	+5.45%	-0.22	0.35

A basket substitution changes every cross-sectional rank and hence other coins' positions; trend models retain fixed per-slot budgets. This is a matched strategy comparison, not a causal coin-removal experiment. Validation-profitability of a control is evidence to retain and investigate, not permission to silently promote it.

3. The basket comparison also changes risk

The same-pair comparison must retain turnover and drawdown, not just returns and Sharpe. These rows use the identical strategy settings, dates, 20 bps fee convention and idealized observed-close execution as the preceding table.

Annual TO is full-L1 target-change turnover in times per year. Maximum drawdown is the positive loss magnitude from the saved continuous two-year path. This table is a comparison of realized paths, not a risk-adjusted portfolio or a causal basket treatment.

Reference risk was not uniformly lower. A03 reversal had a larger validation drawdown on the reference basket (**53.46%**) than on the selected basket (**43.11%**), despite slightly lower turnover. A04's reference drawdown was lower, but still **58.37%**.

Same settings, selected versus reference risk

ID	Period	Selected annual TO	Reference annual TO	Selected max DD	Reference max DD
A01	Development	31.44×	32.69×	23.25%	35.33%
A01	Validation	35.15×	33.52×	44.82%	30.83%
A02	Development	49.58×	52.75×	58.37%	44.13%
A02	Validation	55.43×	50.74×	47.29%	22.08%
A03	Development	27.62×	24.00×	19.79%	32.31%
A03	Validation	22.34×	21.41×	43.11%	53.46%
A04	Development	50.50×	54.25×	51.35%	56.16%
A04	Validation	63.91×	65.29×	70.96%	58.37%

4. Strong development paths depended on period-specific positions

The four candidates were profitable in both development years by construction. Validation exposed a different pattern: A01 recovered in 2025 after a deep 2024 loss; A02–A04 lost in both validation years. Two-year windows and annual views answer different questions, so they are shown together rather than moving window boundaries after seeing results.

Annual net returns on continuous paths

ID	2022	2023	2024	2025
A01	+66.26%	+59.60%	-39.15%	+15.77%
A02	+140.95%	+46.47%	-32.43%	-10.66%
A03	+32.05%	+24.48%	-1.91%	-20.80%
A04	+83.54%	+125.69%	-13.10%	-37.06%

Legacy LUNA slot: saved net-P&L attribution

ID	Development LUNA contribution (pp)	Validation LUNA contribution (pp)
A01	+49.39	-25.46
A02	+132.19	-19.51
A04	+78.96	-19.88

The legacy LUNA slot contributed **+49.39, +132.19 and +78.96 percentage points** to A01, A02 and A04's development net returns, respectively, but negative contributions in validation. These are NAV-weighted accounting contributions to starting capital, not standalone coin returns. The table reports percentage-point (pp) contributions.

All four selected baskets also lost **before modeled fees** in validation: gross compounded returns ranged from **−15.03% to −29.34%**. Costs deepened the losses rather than explaining them in full.

The diagnostic supports **concentration and period-dependence concerns**. It does not establish that deleting LUNA would have produced the same remaining positions or returns. Documented halts, forced exits and symbol lifecycle treatment were preserved; no event was removed to improve the backtest.

5. Relative rank transfer was stronger than economic transfer

The full grid retained **1,117,935 basket–configuration pairs**: 6,615 baskets × 169 settings. Development-ranked groups were frozen before full-grid validation; the 10 previously accepted case-baskets were reused, not resimulated.

Q1 retained a higher validation median Sharpe than lower-ranked groups, but its median net return remained negative. Much of the rank hierarchy reflects the persistent cost burden of fast settings, not proof that a positive-alpha winner transfers.

These medians are constituent distributions, **not portfolios**. A further 1,000 development all-cash cases had undefined Sharpe and remain unranked in the archive rather than being assigned to the worst group.

Frozen development-ranked groups

2022–2023 development

Group	Cases	Median net	Median Sharpe	Median annual TO	Positive fraction
Q1 (best)	224,867	-1.55%	0.14	60.01×	47.19%
Q2	223,954	-37.25%	-0.74	70.25×	0.08%
Q3	219,294	-69.06%	-1.77	159.44×	0.00%
Q4	223,910	-92.90%	-4.03	487.59×	0.00%
Q5 (worst)	224,910	-99.9973%	-15.69	2753.01×	0.00%

2024–2025 validation

Group	Cases	Median net	Median Sharpe	Median annual TO	Positive fraction
Q1 (best)	224,867	-16.27%	-0.15	64.77×	30.73%
Q2	223,954	-35.59%	-0.76	69.73×	12.99%
Q3	219,294	-59.56%	-1.63	162.09×	2.68%
Q4	223,910	-90.15%	-4.18	489.30×	0.01%
Q5 (worst)	224,910	-99.9977%	-17.87	2654.28×	0.00%

6. A high rank need not imply a positive return

Across the full grid, **62.90% of strictly positive development results became strictly negative** in validation. Conversely, **6.40% of strictly negative development results became positive**. The denominators differ, so these are conditional transition rates rather than symmetric probabilities. Exact zeros remain separate.

The best development-ranked group had **224,867 cases**; **69,108 (30.73%)** were profitable in validation. Of these, **39,332** also had positive development returns. The remaining **29,776** were development-negative cases that survived validation. These counts are not independent replications.

How often the return sign changed

Family	Dev positive	Positive → negative	Dev negative	Negative → positive
All families	106,302	62.90%	1,010,633	6.40%
Momentum	49,083	72.29%	466,887	8.69%
Reversal	13,497	90.91%	502,473	1.79%
Buffered momentum	10,719	66.24%	15,741	35.04%
Long/cash trend	14,898	18.53%	11,562	63.81%
Long/short trend	17,355	49.43%	9,105	24.91%
BTC residual reversal	750	90.40%	4,865	0.00%

Development–validation Sharpe rank correlation

Comparison	Cohorts	Median Spearman	10th percentile	90th percentile
Across settings within basket	6,615	0.91	0.86	0.94
Across baskets within setting	169	0.14	-0.23	0.62

The median rank correlation was **0.91 across settings within a basket**, but only **0.14 across baskets within the same setting**. That distinction explains why broad setting hierarchies can persist while the exact development-winning basket is unstable.

No IID p-value or confidence interval is attached to these full-grid cross-sectional summaries. Adjacent baskets share seven coins; configurations share signals, prices and dates. Long/cash trend's lower sign-flip rate is descriptive, not a new final-test endorsement.

7. Track B was an honest validation-informed selection

After validation had been inspected, a separate rule was frozen **before executing its selection**. Start with development Q1, require evaluated statuses, finite defined Sharpes and strictly positive net returns in both development and validation at 20 bps.

Within each of the existing six families, select the highest validation Sharpe. Ties use lower validation full-L1 turnover, lexical experiment ID and numeric basket ID. Select none when no case qualifies. No positive-Sharpe threshold, annual consistency, stress-cost survival, neighborhood success or risk cap was added as a hidden eligibility gate.

Every family was retained, including the empty one

ID	Family	Full grid	Dev Q1	Eligible	Selected
B01	Buffered momentum	26,460	20,068	3,619	1
B02	BTC residual reversal	6,615	1,355	0	0
B03	Momentum	515,970	107,412	13,585	1
B04	Reversal	515,970	45,673	1,227	1
B05	Long/cash trend	26,460	24,856	12,125	1
B06	Long/short trend	26,460	25,503	8,776	1

B02 was never a failed selected strategy

B02 is the reserved family label for BTC-factor residual reversal. **Zero** of its **1,355** development-Q1 pairs met the positive-in-both-periods rule, so no B02 candidate was selected. The earlier factor reference control was a diagnostic comparison, not a Track B selection. An empty family is a valid result; it is not replaced by another family or treated as a zero-return investment.

All five selected validation Sharpes are selection statistics from a large correlated pool. They are not unbiased final-test Sharpe estimates, and do not retroactively repair Track A.

8. Track B's strong validation results came with weak development cases

B01 and B03 barely cleared the positive-development-return gate. They were eligible under the frozen rule; annual consistency was diagnostic only. Their validation improvement deserves investigation without pretending it was forecast by strong development performance.

Frozen Track B cases, unchanged in each period

2022–2023 development

ID	Net return	Sharpe	Annual L1 turnover	Max drawdown
B01	+0.34%	0.14	81.69×	37.34%
B03	+1.05%	0.15	27.75×	27.64%
B04	+44.11%	0.63	21.31×	61.98%
B05	+12.60%	0.35	11.88×	36.50%
B06	+35.86%	0.55	22.88×	48.23%

2024–2025 validation: selection statistics

ID	Net return	Sharpe	Annual L1 turnover	Max drawdown
B01	+120.19%	1.62	77.45×	15.50%
B03	+151.30%	1.85	24.13×	22.08%
B04	+73.78%	1.14	22.34×	23.42%
B05	+152.00%	1.40	12.48×	32.48%
B06	+164.24%	1.24	24.59×	32.83%

Annual consistency was not a Track B gate

ID	2022	2023	2024	2025
B01	+8.87%	-7.84%	+48.93%	+47.85%
B03	-9.01%	+11.06%	+26.72%	+98.31%
B04	+36.70%	+5.42%	+46.66%	+18.49%
B05	-33.96%	+70.51%	+111.32%	+19.25%
B06	+8.59%	+25.11%	+56.17%	+69.20%

9. Nine exact identities were frozen for evaluation

BTC and ETH are fixed in every basket; the table lists the other six slots. h is trailing lookback hours, r is rebalance interval hours, and b025 is the whole-target full-L1 buffer of 0.25. Weekly r168 decisions occur Monday at 00:00 UTC; r072 uses the established elapsed-time schedule.

Track A remains the original development-selected procedure; Track B remains the development-plus-validation selection. The shortlist is **not a portfolio**, has no ensemble weights, and does not designate a winner.

Case–basket identity register

Track A: four original selections

ID	Strategy	Basket	Other six coins
A01	buffered_momentum_h0672_r168_b025	4,645	SOL, LUNA, FTM, TRX, LINK, LTC
A02	momentum_h0672_r072	4,645	SOL, LUNA, FTM, TRX, LINK, LTC
A03	reversal_h1440_r168	1,159	SHIB, DOT, ADA, MANA, FIL, VET
A04	trend_long_short_h0672_r024	4,444	SOL, LUNA, AVAX, FTM, FIL, VET

Track B: five validation-informed selections

ID	Strategy	Basket	Other six coins
B01	buffered_momentum_h0672_r024_b025	2,323	BNB, XRP, DOGE, FTM, AXS, VET
B03	momentum_h2160_r072	2,457	BNB, XRP, FTM, MATIC, AXS, LTC
B04	reversal_h2160_r168	6,064	DOT, LUNA, AVAX, MANA, FIL, VET
B05	trend_long_cash_h0672_r168	2,330	BNB, XRP, DOGE, TRX, GALA, LINK
B06	trend_long_short_h0672_r168	2,495	BNB, XRP, TRX, MANA, GALA, LINK

10. What was frozen, and what these results can support

Do not move windows after inspecting performance. Development and validation each span two complete years, chosen before this final editorial consolidation. Annual slices inherit the continuous path's positions and fees; there is no invented internal-year liquidation or cash reset.

Report the right Sharpe. Development Sharpe describes fitted evidence; Track A validation Sharpe describes transfer of a development-selected rule; Track B validation Sharpe is selection-biased. The subsequent evaluation must retain that selection history and the earlier project's inspection of 2026. Freezing a restart does not make already-seen history pristine.

Separate statistical aims. Track A asks whether a development-selection procedure generalizes. Track B asks whether a strategy selected using development and validation transfers to a later period. Neither requires the development winner to remain rank one, but economic and risk expectations must be stated before that later evaluation.

Reserved evaluation protocol. The nine identities and approved risk limits remain fixed. The later approved amendment uses 1 January–30 September 2026 (273 days), rather than a complete calendar year. It requires daily-mean t-statistics, block-bootstrap intervals, multiple-testing correction and honest partial-year labels. No 2026 results are reproduced in this report.

Common measurement and limitations

Net return compounds the saved hourly gross P&L less one-way fee × full-L1 target change, including actual entry and terminal liquidation. Annual turnover is 365 × mean daily full-L1 target change, **not** round-trip turnover or drift-adjusted executed notional. Maximum drawdown is positive peak-to-trough loss magnitude; annualized daily Sharpe uses sample SD, 365 days and zero risk-free rate.

The sensitivity set is 0/5/10/15/40 bps around the 20 bps primary scenario, plus 20 bps with one additional execution hour. Actual borrowing, funding, margin, order-book depth, impact and ordinary drift-maintenance trading were not modeled. Spot shorts are theoretical; historical liquidity and surviving economic slots do not remove universe or lifecycle bias.

This publication edition consolidates accepted saved metrics and diagnostics only. Source menus expose each component's exact evidence. Detailed configurations, failed cases, cost scenarios, annual risk tables, model logs and provenance remain in the unchanged archival studies; they are supporting evidence, not additional final reports.